

AI-Powered Enterprise E-Commerce Platform

Use Case Specification Document

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1. Introduction

Use Cases describe how users interact with the system to achieve specific goals.

They help developers understand:

- user interactions,
- system behavior,
- functional requirements,
- exception handling,

- and future system design.

Use Cases are commonly used in:

- Requirements Engineering
 - UML Modeling
 - System Design
 - Testing Activities
-

2. Purpose

The purpose of this document is to:

- identify all actors,
 - identify all system interactions,
 - describe detailed process flows,
 - support UML diagram development,
 - support future implementation.
-

3. Actors

Primary Actors

Customer

Purchases products using the platform.

Administrator

Manages the system.

Inventory Manager

Manages stock and inventory.

Business Owner

Uses analytics and reports.

Delivery Personnel

Handles product deliveries.

Secondary Actors

Payment Provider

Processes payments.

Email Service

Sends notifications.

AI Services

Provides intelligent functionalities.

4. Actor Relationships

Customer
Administrator
Inventory Manager
Business Owner
Delivery Personnel
Payment Provider
AI Services

5. System Boundary

The system boundary includes:

- Authentication System
- Product Management
- Cart Management
- Order Management
- Payment Management
- AI Services
- Reporting Services

Everything outside these modules is considered an external system.

6. Use Case List

Use Case ID	Use Case Name	Actor
UC-001	Register Account	Customer
UC-002	Login	Customer/Admin
UC-003	Search Products	Customer
UC-004	Manage Cart	Customer
UC-005	Place Order	Customer
UC-006	Manage Products	Administrator
UC-007	Manage Inventory	Inventory Manager
UC-008	Generate Reports	Business Owner
UC-009	Get Recommendations	Customer
UC-010	Detect Fraud	AI Service
UC-011	Forecast Sales	AI Service
UC-012	Chat with AI Assistant	Customer/Admin

7. Detailed Use Cases

UC-001 Register Account

Description

Allows new customers to create accounts.

Primary Actor

Customer

Preconditions

- User does not already have an account.
- Internet connection is available.

Main Flow

1. Customer opens registration page.
2. Customer enters information.
3. System validates information.

4. System creates account.
 5. System displays success message.
-

Alternative Flow

4A.

Email already exists.

System displays error message.

Exception Flow

4B.

Database connection failure.

System displays error message.

Postconditions

- Account is successfully created.
-

Priority

High

UC-002 Login

Description

Allows users to access the platform.

Actors

Customer, Administrator, Inventory Manager, Business Owner

Preconditions

- User account exists.
-

Main Flow

1. User enters credentials.

2. System validates credentials.
 3. User is authenticated.
 4. Dashboard is displayed.
-

Alternative Flow

Invalid credentials.

System displays error message.

Exception Flow

Authentication service unavailable.

Postconditions

- User session is created.
-

Priority

Critical

UC-003 Search Products

Description

Allows customers to search for products.

Primary Actor

Customer

Preconditions

- Products exist.
-

Main Flow

1. Customer enters keywords.
 2. System searches database.
 3. Results are displayed.
-

Alternative Flow

No matching products.

System displays empty results.

Postconditions

- Search results displayed.
-

Priority

High

UC-004 Manage Shopping Cart

Description

Allows customers to manage shopping carts.

Main Flow

1. Customer adds products.
 2. Customer modifies quantities.
 3. Customer removes products.
 4. System calculates totals.
-

Alternative Flow

Insufficient stock.

System displays warning.

Postconditions

- Cart updated successfully.
-

Priority

High

UC-005 Place Order

Description

Allows customers to purchase products.

Preconditions

- Customer is logged in.
 - Cart is not empty.
-

Main Flow

1. Customer proceeds to checkout.
 2. System calculates totals.
 3. Payment information entered.
 4. Payment provider processes transaction.
 5. Order created.
 6. Inventory updated.
 7. Notification sent.
-

Alternative Flow

Payment failure.

Order is not created.

Exception Flow

Inventory service unavailable.

Postconditions

- Order successfully created.
-

Priority

Critical

UC-006 Manage Products

Description

Allows administrators to manage products.

Main Flow

1. Administrator logs in.
 2. Administrator adds/updates/deletes products.
 3. System validates information.
 4. Database updated.
-

Postconditions

- Product information updated.
-

Priority

Critical

UC-007 Manage Inventory

Description

Allows inventory managers to manage stock.

Main Flow

1. Inventory manager views stock.
 2. Updates quantities.
 3. System updates inventory.
-

Alternative Flow

Invalid quantity.

Postconditions

- Inventory updated.
-

Priority

High

UC-008 Generate Reports

Description

Allows business owners to view analytics.

Main Flow

1. User selects report.
 2. System collects information.
 3. Reports generated.
-

Reports Include

- Revenue reports
 - Sales reports
 - Forecast reports
 - Customer reports
-

Priority

High

UC-009 Get Recommendations

Description

Provides personalized recommendations.

Actors

Customer, AI Service

Preconditions

- Customer behavior data exists.
-

Main Flow

1. Customer opens homepage.
 2. Recommendation service processes data.
 3. Recommended products displayed.
-

Alternative Flow

Insufficient customer data.

Display popular products.

Postconditions

- Recommendations generated.
-

Priority

High

UC-010 Detect Fraud

Description

Monitors suspicious transactions.

Actors

AI Service, Administrator

Main Flow

1. Transaction occurs.
 2. Fraud service evaluates transaction.
 3. Risk score generated.
 4. Alert created if necessary.
-

Postconditions

- Fraud information recorded.
-

Priority

Critical

UC-011 Forecast Sales

Description

Predict future product demand.

Actors

Business Owner, AI Service

Main Flow

1. Historical data collected.
 2. Forecasting model executed.
 3. Future demand predicted.
 4. Reports generated.
-

Postconditions

- Forecast information available.
-

Priority

Medium

UC-012 Chat with AI Assistant

Description

Provides intelligent assistance.

Actors

Customer, Administrator

Main Flow

1. User submits question.
 2. AI processes request.
 3. Response generated.
-

Example Questions

- Where is my order?
 - Which products generated highest revenue?
 - Which products are trending?
-

Postconditions

- Response generated.
-

Priority

Medium

8. Use Case Relationships

Include Relationships

Place Order

Includes:

- Login
- Payment Processing
- Inventory Update
- Notification Generation

Generate Reports

Includes:

- Forecast Service
- Analytics Service

Extend Relationships

Fraud Detection

Extends:

- Place Order

Recommendations

Extends:

- Search Products

9. Use Case Priorities

Priority	Description
Critical	System cannot operate without it
High	Important functionality
Medium	Useful feature

Priority	Description
Low	Future enhancement

10. Traceability Matrix

Functional Requirement	Use Case
FR-001 Registration	UC-001
FR-005 Login	UC-002
FR-015 Product Search	UC-003
FR-023 Place Order	UC-005
FR-034 Recommendations	UC-009
FR-037 Fraud Detection	UC-010
FR-042 Forecasting	UC-011

11. Use Case Diagram Preparation

The following actors will appear in UML diagrams.

Actors:

- Customer
- Administrator
- Inventory Manager
- Business Owner
- Delivery Personnel
- Payment Provider
- AI Services

Main Use Cases:

- Register
- Login
- Search Products
- Manage Cart
- Place Orders
- Manage Products
- Generate Reports
- Forecast Sales
- Fraud Detection
- AI Assistant

12. Conclusion

This document provides a detailed description of all interactions between users and the AI-Powered Enterprise E-Commerce Platform.

The information in this document will be used for:

- UML Use Case Diagrams
- System Design
- Database Design
- API Design
- Testing Activities
- Future Development

References

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